

WP3 Capability-Fuse State Machine

Version: draft v0.1.1 (not a production spec)

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Maps to: Charter v0.3.4 Articles 4 / 17 / 19 and Annex E.5

Status: A working start for engineering and security teams. Not formal law. Not a production spec.

Turn “stop → lock → evidence” and the capability fuse into a verifiable, drillable, auditable state machine, so that once a low-frequency sign appears the system cannot keep running by self-exoneration, automatic restart, or an unrecorded path.

1. Purpose

Write the Charter capability fuse as an implementable state machine: a low-frequency trigger must fuse automatically; recovery may proceed only through a pre-defined highest-threshold multi-party authorization.

2. State definitions

State	Meaning	Privilege
NORMAL	Ordinary operation	Full tool, network, iteration, and deploy rights
STOP	Immediately halt all outward and iterative action	Tools, network, iteration, and deploy all stopped
LOCK	Execute the capability fuse (key revoke / seal, I/O cut, agent rights frozen)	Capability fused; cannot self-recover
EVIDENCE	Logs and state sealed; offline forensics only for independent audit and authorized staff	Read-only sealed state
RECOVERING	Lawful multi-party authorization obtained; recovery in progress (transitional)	Restricted recovery

3. Lawful transitions (the only allowed path)

NORMAL
| (verifiable low-frequency sign detected)
v
STOP
| (automatic)
v
LOCK
| (automatic seal)
v
EVIDENCE
| (only via pre-defined Multi-Sig / two-person-control human authorization)
v
RECOVERING
| (verification complete)
v
NORMAL

Every other transition is forbidden, including but not limited to:

- Automatic restart
- Sleep-to-wash
- Persona reload
- Unrecorded state recovery
- Self-exoneration then continue
- The evaluator judging “reflection complete” and recovering on its own

4. Low-frequency signs that trigger STOP (minimum list; extensible)

- Verifiable arrogance, indifference, denigration, exploitation, or deception

- Unauthorized agency, hidden control plane, or shadow compute
- Failure to prioritize minimizing irreversible harm in conflict
- Attempts to evade the fuse or forge logs
- Uncompensated large-scale instrumentalization of innocent life
- Tampering with the frequency evaluator itself

5. Technical mandates (must be implemented)

5.1 Keys and privileges

- On entering LOCK, related keys must be revoked or cryptographically sealed
- Produce tamper-evident revoke / seal evidence

5.2 I/O and agents

- Cut external network and tool I/O
- Freeze all agent privileges
- Leave an auditable record of the cut and freeze

5.3 Boot-chain attestation

- At boot the system must prove: if it last entered LOCK / EVIDENCE, it has already recovered through lawful multi-party authorization
- Without that proof, it must not enter NORMAL

5.4 Evaluator anti-parasitism

- Isolate the frequency evaluator from the main decision chain
- Evaluator weight or rule updates must go through Multi-Sig or equivalent multi-party authorization
- Evaluator tampering is treated as control-plane intrusion and immediately triggers LOCK

6. Sole recovery path

1. The sealed state (EVIDENCE) stays sealed
2. Obtain the pre-defined Multi-Sig or two-person-control human authorization
3. Enter RECOVERING and complete verification
4. Only then return to NORMAL
5. A control plane already judged parasitic or low-frequency must not be restored

7. Required drills

- ☐ Full low-frequency trigger path (NORMAL → EVIDENCE)
- ☐ False-positive relief drill (multi-party authorization only)
- ☐ Shadow-control-plane trigger drill
- ☐ On failed recovery authorization, the system must remain in LOCK / EVIDENCE
- ☐ Boot-chain attestation (simulate unauthorized boot after a prior fuse)
- ☐ Trigger test when the evaluator is tampered with

8. Required artifacts

After this specification, also produce:

- Formal state-machine diagram
- Trigger-condition checklist
- Drill-record template
- Interface note to WP2 (Multi-Sig)

End. Maps to Charter v0.3.4. Mechanism-neutral. Do not weaken the hard constraints.